

IECEx/ATEX Certification Information Requirements

1. Basic Information (Application and Organization):

- 1) Application form (provided by ExCB as a template).
- 2) Manufacturer's business license/company registration certificate.
- 3) Manufacturer's organizational chart and contact information.
- 4) Address of manufacturing plant and brief introduction of production facilities.
- 5) If applicable) ISO 9001 certificate, quality manual, and directory of procedure documents (for QAR/QAN review).

2. Technical Documentation:

- 1) Detailed product description;
- 2) Product name, model, and specification list;
- 3) Complete and clear functional description and technical parameters (electrical parameters, mechanical parameters, etc.);
- 4) Clear product photos (overall and key components);
- 5) Description of intended use and installation environment (e.g., gas group IIC/IIB/IIA if applicable, temperature group T1-T6, ambient temperature range, area classification 0/1/2 or 20/21/22);
- 6) Design and manufacturing information:
 - a. Complete drawings: Assembly drawing, component drawing, electrical schematic diagram, PCB layout (if applicable), enclosure structure drawing (indicating key explosion-proof parameters such as gap, length, volume), junction box drawing, terminal drawing, grounding connection drawing, etc. Drawings should clearly indicate dimensions, tolerances, and materials;
 - b. Bill of Materials (BOM) for key components: List all components that affect explosion-proof safety (such as enclosures, gaskets, terminals, cable entry devices, switches, light sources, batteries, potting compounds, plastic parts, printed circuit boards, etc.), including manufacturer, model and specification, material grade (e.g., in compliance with IEC 60079-0 Appendix A), certification information (e.g., existing UL, ATEX, CCC certificates, etc.) or technical parameters (such as CTI value, flame retardancy, chemical resistance, glass transition temperature T_g, etc.);
 - c. Material specification sheets/data sheets: Detailed technical parameters for non-metallic materials (enclosures, seals, insulators, potting compounds, etc.) to prove compliance with standard requirements (such as heat and cold resistance, anti-aging, anti-static, flame retardancy, CTI value, comparative tracking index, chemical resistance, etc.). Provide test reports or conformity declarations from material suppliers;
 - d. Manufacturing process description: Especially for processes that affect explosion-proof safety (such as enclosure processing accuracy control, explosion-proof surface treatment, welding process, potting process, die-casting process, bonding process, etc.);
 - e. Explosion-proof related calculations and explanations (if applicable);
 - f. Temperature group calculation (estimation of the hottest point temperature);
 - g. Usage and installation instructions (draft):
 - a) Clear safety warnings;
 - b) Detailed installation requirements (such as cable entry method, torque requirements, grounding requirements, cleaning requirements);
 - c) Operating instructions, maintenance requirements (frequency, content, precautions);
 - h. Nameplate information (must include all ATEX required marking information);
 - i. Ex components ATEX certificates

3. Sample Information:

- 1) Provide the model, serial number, and status (such as whether it can be powered on and run) of the sample;
- 2) Provide necessary test fixtures or auxiliary equipment.

IECEX/ATEX 认证信息提供要求

1. 基本资料 (申请与组织):

- 1) 申请表 (由ExCB提供模板)。
- 2) 制造商营业执照/公司注册证明。
- 3) 制造商组织结构图、联系信息。
- 4) 制造工厂地址、生产设施简介。
- 5) (如适用) ISO 9001 证书及质量手册、程序文件目录 (用于QAR评审)。

2. 技术文件 (核心):

- 1) 产品详细描述:
- 2) 产品名称、型号、规格清单。
- 3) 完整、清晰的功能描述和技术参数 (电气参数、机械参数等)。
- 4) 清晰的产品照片 (整体和关键部件)。
- 5) 预期用途和安装环境描述 (如适用气体组别IIC/IIB/IIA, 温度组别T1-T6, 环境温度范围, 区域划分0/1/2或20/21/22)。
- 6) 设计与制造信息:

- a. 完整的图纸：总装图、部件图、电路图、电气原理图、PCB布局图（如适用）、外壳结构图（需标明关键隔爆参数如间隙、长度、容积）、接线盒图、端子图、接地连接图等。图纸需清晰标注尺寸、公差、材料。
 - b. 关键部件清单 (BOM)：列出所有影响防爆安全的零部件（如外壳、衬垫、端子、电缆引入装置、开关、光源、电池、灌封胶、塑料件、印制电路板等），包括制造商、型号规格、材料等级（如符合IEC 60079-0附录A）、认证信息（如已有UL, ATEX, CCC等证书）或技术参数（如CTI值、阻燃等级、耐化学性、玻璃化温度 T_g 等）。
 - c. 材料规格书/数据表：特别是非金属材料（外壳、密封件、绝缘件、灌封胶等）的详细技术参数，证明其符合标准要求（如耐热耐寒、抗老化、抗静电、阻燃性、CTI值、相比漏电起痕指数、耐化学腐蚀性等）。提供材料供应商的测试报告或符合性声明。
 - d. 制造工艺描述：特别是对防爆安全有影响的工艺（如外壳加工精度控制、隔爆面处理、焊接工艺、浇封工艺、压铸工艺、粘接工艺等）。
 - e. 防爆相关计算与说明 (如适用)：
 - f. 温度组别计算（最热点温度估算）。
 - g. 使用与安装说明 (草案)：
 - a) 清晰的安全警告。
 - b) 详细的安装要求（如电缆引入方式、扭矩要求、接地要求、清洁要求）。
 - c) 操作说明,维护要求（周期、内容、注意事项）。
 - h. 铭牌信息（需包含所有IECEx要求的标志信息）。
 - i. 防爆元器件ATEX证书
- 7) 样品信息：
- a. 提供样品的型号、序列号、状态（如是否可通电运行）。
 - b. 提供必要的测试工装或辅助设备。